

Machine Learning with Orange

Session 1: Classification & Regression

— Data → Preprocess → Model → Evaluate —

What is Machine Learning?



Learn from Data

Systems improve from examples, not explicit rules



Find Patterns

Discover structure hidden in raw numbers



Make Predictions

Apply learned patterns to new, unseen data

ML = Teaching computers to learn from experience



Data

Collect & load your dataset



Preprocess

Clean & prepare features



Model

Train algorithm to learn patterns



Evaluate

Test performance & validate

Step 1: Data

- 📁 Rows = instances (samples)
- ☰ Columns = features (attributes)
- 🏷️ Target = what we want to predict

Student	GPA	Attendance	Study_Hours	Result
A	3.5	85%	8	Pass
B	2.8	60%	4	Fail
C	3.9	90%	10	Pass

In Orange: use File widget or built-in datasets

Step 2: Preprocessing — Why Preprocess?

Missing Values

Fill or remove?
Use mean, median, or drop rows

Feature Scaling

Normalize (0–1)
or
Standardize (z-score)

Feature Selection

Keep only relevant columns

In Orange: Impute, Normalize, Select Columns widgets

Step 3: Model — Two Types of Supervised Learning



CLASSIFICATION

Predict a category

Examples: Spam/Not Spam, Disease/Healthy, Pass/Fail



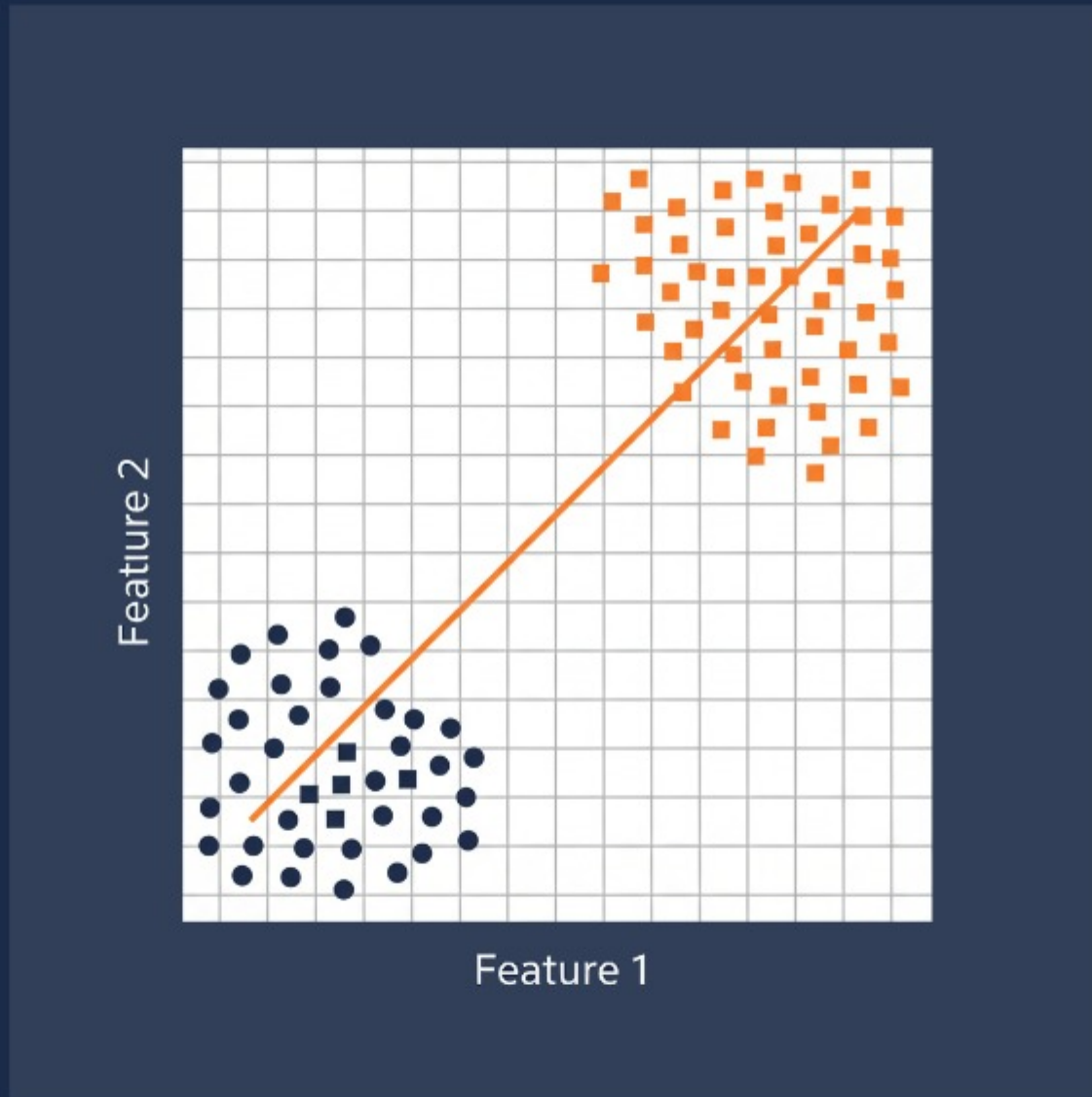
REGRESSION

Predict a number

Examples: House price, Temperature, Student score

The difference: Is the target a label or a number?

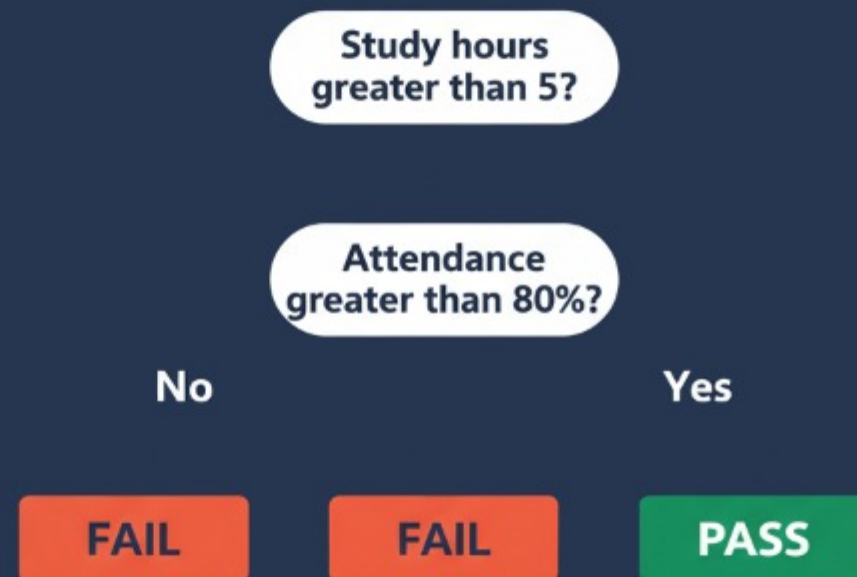
Classification — How It Works



- Model learns a boundary between classes
- New data point → assigned to nearest class
- Example algorithms: Decision Tree, k-NN, Naive Bayes

Think of it as: drawing lines that separate groups

Decision Tree – Intuition

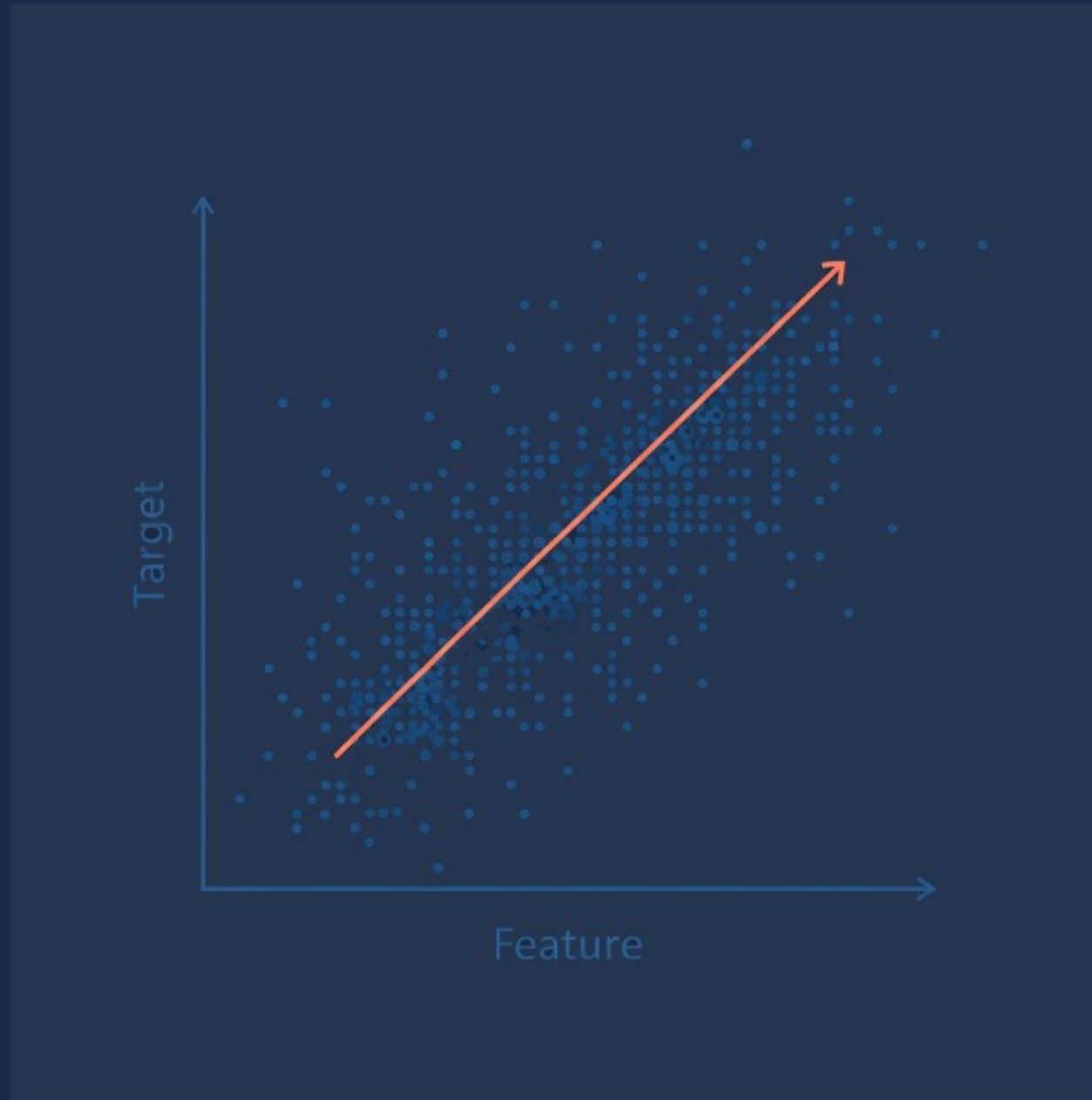


Key Properties

- Easy to interpret
- Works with categorical & numeric data
- Risk: overfitting on small data

In Orange: Tree widget + Tree Viewer for visualization

Regression — How It Works



- Model learns a function: input features $\rightarrow$ continuous output
- Loss function measures prediction error
- Example algorithms: Linear Regression, kNN Regression

Think of it as: fitting the best line through data points

Step 4: Model Evaluation — How Do We Know the Model is Good?

70% Train

30% Test

Classification Metrics

- Accuracy — % correct predictions
- Precision & Recall — for imbalanced classes
- Confusion Matrix — see where model fails

Regression Metrics

- MAE — average absolute error
- RMSE — penalizes large errors more
- R^2 — how much variance is explained

Confusion Matrix — Don't Be Confused

	Predicted Positive	Predicted Negative
Actual Positive	TP True Positive	FN False Negative
Actual Negative	FP False Positive	TN True Negative

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total}$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

In Orange: Confusion Matrix widget gives this automatically

Cross-Validation — Why Not Just Train/Test Once?



- Single split can be lucky or unlucky
- k-Fold CV: average performance across k splits
- More reliable estimate of real-world performance

In Orange: Test & Score widget handles this for you

Putting It All Together in Orange



Today's lab: replicate this workflow with a real dataset

Session 2 — Key Takeaways

- 1 ML learns patterns from data — no manual rules needed
- 2 Always preprocess: handle missing values & scale features
- 3 Classification predicts labels; Regression predicts numbers
- 4 Evaluate with proper metrics — accuracy alone can be misleading

Next session → Clustering & Time Series